[METRIC]
H-B-141K
March 20, 1995
Superseding
H-B-141J
November 15, 1991

FEDERAL SPECIFICATION

BRUSH, CALCIMINE, WALL COATING

The General Services Administration has authorized the use of this federal specification for all federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers brushes used for coating large wall areas.

1.2 Classification. The brushes shall be of the following types, grades, and sizes as specified (see 6.2): (Size denotes the width of brush inside the ferrule.)

TYPE I - Dutch (Block Style), 150 mm. (6 in.)

Class 1 - Bristle — Class 2 - Synthetic Filament

TYPE II - Flat, 150 mm. (6 in.)

Class 1 - Bristle — Class 2 - Synthetic Filament

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-1880 - Brushes; Preparation for Delivery of
H-S-951 - Synthetic Filaments for Brushes, General Spec For

(Copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available from the following address: GSA Specifications Section (3FBP-W), Suite 8100, 470 L'enfant Plaza SW, Washington, DC 20407.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Beneficial comments, recommendations, additions, deletions, clarification, etc., and any data which may improve this document, should be sent to: GSA, Paints and Chemicals Commodity Center, 400 15th Street, S.W., Auburn, WA 98001.

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2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply:

American Society for Testing and Materials:

ASTM D13 – Specification for Spirits of Turpentine

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Materials and Components. Materials and components shall conform to the requirements specified herein.

3.1.1 Bristle (Class I). The bristle for Class I brushes shall be unbleached and undyed natural hog bristle. The bristle shall be boiled or steamed and straightened so that it will not twist or curl, and shall be free from reconditioned bristle or adulterants. The bristle shall pass the straightness test specified in 4.4.4. A certificate of compliance from the supplier that the bristle meets all requirements (stiffness and sterilization) may be accepted. The contractor shall provide certification that color(s) has not been added to the bristles.

3.1.2 Synthetic filaments (Class II). The synthetic filament for Class II brushes shall be organic polymeric material conforming to H-S-951, class N, type I, or Class E, Type I, tapered filaments meeting the size designation of Table II. All filaments shall be tipped to a depth of not less than 0.48 cm. (3/16 in.). After tipping, 20 percent by weight of the top-sized filaments shall be flagged. (See 6.3.1)

3.1.3 Ferrule. The ferrule shall be made from corrosion-resistant metal or metal made corrosion-resistant by electro-plating.

3.1.4 Handle. The handle shall be either a smoothly finished closed-grained hardwood or a synthetic material with all sprues and runners removed. The synthetic handle shall not crack or break when tested as in 4.4.3. The handle organic coating material shall not wrinkle, blister, or separate or have any deleterious effect when tested as in 4.4.2.1. Synthetic handle base material shall not dull or soften when tested as in 4.4.2.2.

3.1.5 Filler strips. Filler strips shall be made of wood, metal, or plastic and shall not soften or otherwise change when tested as in 4.4.2.2.

3.1.6 Setting compound. The setting compound shall be an epoxy cement which shall withstand the test specified in 4.4.2.2 without softening or deteriorating, and without loosening of bristle in the compound.

3.2 Construction

3.2.1 Ferrule. The ferrule shall have a minimum wall thickness of .3 mm. (0.010 in.). The ferrules shall be formed with an interlocked, lock-seamed, soldered, or welded end lap, and shall have a groove at the base to aid in controlling the flow of the setting compound. The ferrule height shall be 25 mm. (31/32 in.) minimum for
Type I brushes and 31 mm. (1 7/32 in.) minimum for Type II brushes.

3.2.2 Set. The butt ends of the bristles or filaments shall be set in a setting compound conforming to 3.1.6. When tested as in 4.4.2.2, there shall be no more than a total of five (5) bristles or filaments pulled loose during the tensile pull.

3.2.3 Assembly. The ferrule shall be fastened to the handle with 17 gage minimum corrosion-resistant nails 10 mm. (3/8 in.) minimum in length. There shall be a minimum of four (4) evenly spaced nails on each side of the brush.

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orm Packing - The packing shall be uniform in texture and shall pert

3.2.4 Trim. The brushes shall be formed to a square top with the tipped or flagged ends preserved.

3.3 Composition and physical requirements.

3.3.1 Class I. The bristle shall be 60 percent top size and blended in accordance with Table I. (Dressed bristle, described as 60 percent top size, denotes that bundle consists of 60 percent bristle measuring 2 mm. (1/16 in.) or more above the length specified to 3 mm. (1/8 in.) below the length specified; 30 percent to consist of bristle from 3 mm (1/8 in.) to 10 mm. (3/8 in.) below the length specified; 10 percent will be 10 mm. (3/8 in.) to not more than 13 mm. (1/2 in.) below specified length, but including a few unavoidable shorts.) No less than 40 percent nor more than 60 percent of the bristles shall be stiff. Bristles shall be blended as specified in Table I.

TABLE I. BLEND OF BRISTLE

Length of bristle Size, mm. (in)	Type I Percent	Type II Percent
140 (5-1/2)	–	80
133 (5-1/4)	–	10
127 (5)	80	5
121 (4-3/4)	10	5
114 (4-1/2)	5	–
108 (4-1/4)	5	–

3.3.2 Synthetic filament for class 2 brushes. The synthetic filament shall be of the blend specified in table II and thoroughly mixed. A tolerance of plus or minus 3 percent will be permitted for each length specified.

TABLE II. BLEND OF FILAMENT

Filament size designation	Length of filament, mm. (in.)			
	121 (4-3/4)	114 (4-1/2)	108 (4-1/4)	102 (4)
12 - 8	40 %	–	–	–
15 - 10	15 %	15 %	15 %	15 %

3.4 Physical Requirements. The brushes shall conform to the requirements in Table III

TABLE III. PHYSICAL REQUIREMENTS

Characteristics	Type I	Type II
Width of brush inside ferrule, mm. (in.), min.	152 (6)	152 (6)
Thickness of brush inside ferrule, mm (in.), min.	38 (1-1/2)	22 (7/8)
Hold of bristles or filament in ferrule, mm. (in.), min.	10 (3/8)	10 (3/8)
Bristle length outside ferrule, mm. (in.), min.	117 (4-5/8)	130 (5-1/8)
Filament length outside ferrule, mm. (in.)	110 (4-5/16)	110 (4-5/16)
Bristle weight per brush, grams (class 1), min.	163	137
Filament weight per brush, grams (class 2), min.	180	170

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3.4 Marking for identification. Unless otherwise specified (6.2), and in addition to any special marking required by the contract or order, each brush shall be stamped or otherwise permanently marked with the manufacturer's name or trademark and the National Stock Number (NSN).

3.5 Workmanship. The finished brushes shall be clean, well made, securely assembled, and free from any defects which may affect appearance or serviceability. There shall be no sharp edges, splinters, burrs, or other imperfections. The brushing part shall be trimmed evenly and uniformly with the flagged ends preserved to the finished length outside the ferrule.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein using facilities acceptable to the Government. The Government reserves the right to perform any inspection herein when deemed necessary to assure that the brushes conform to prescribed requirements.

4.2 Inspection of preparation for delivery. An examination shall be made to determine compliance with 3.3 and section 5 of this specification. The examination shall be conducted as specified in PPP-B-1880.

4.3 Inspection of the end item. Lots shall be assembled and randomly sampled in accordance with MIL-STD 105, inspection level II, with an acceptable quality level (AQL) of 2.5 percent defective. Sample brushes shall be examined for the following attributes:

Bristles or filaments
Ferrules
Handles
Construction
Set
Trim
Workmanship
Identification markings

Any brush in the sample containing one or more visual or dimensional defects shall be considered a defective brush.

4.3.1 Sampling for testing of the end item. A random sample of brushes shall be selected from each lot in accordance with MIL-STD-105, inspection level S-2, with an acceptable quality level (AQL) of 2.5 percent defective.

4.4 Testing procedures. Samples, selected in accordance with 4.3.1, shall be subjected to the following tests. Failure to pass any test shall be cause for rejection of the lot. All tests shall be evaluated for conformance to the requirements in section 3.

4.4.1 Weight of bristle or filament. After removal of the ferrule, the setting compound shall be removed from the bristle by any practical means which does not injure the bristle. The weight of bristles shall be determined for compliance with physical requirements.

4.4.2 Solvent resistance. Solvent resistance tests shall be performed at room temperature.

4.4.2.1 Handle (Organic coating). Brush handles with organic coating shall be tested as follows:

A. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in a commercially available mineral spirits. After 30 minutes, the handle shall be removed and examined. There shall be no wrinkling, blistering, or separation of the coating.
B. With the brushing end up, approximately one-half of the handle shall be immersed for 30 minutes in turpentine conforming to ASTM D13. After 30 minutes, the handle shall be removed from the turpentine and examined. There shall be no wrinkling, blistering, or separation of the coating.

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C. The brush handle shall be dried for 2 hours and the organic coating shall be examined. The coating shall not be wrinkled, blistered, or separated from the handle and the coating shall not be soft.

4.4.2.2 Base material, ferrule, filler strips, synthetic handles. The brush shall be tested for solvent resistance using a commercially available lacquer thinner. (NOTE: The organic coating on the handle may be removed by this solvent. The objective is to test solvent resistance of the handle base material, plating on the ferrule, filler strips, and the setting compound.)

The brush shall be suspended so that the brushing part and approximately one-half of the ferrule are immersed in lacquer thinner for 1 hour. Synthetic handles shall be immersed to a depth of one-half of the handle length. The brush shall be removed and examined as follows:

A. Handle and ferrule - As soon as the brush is removed from the lacquer thinner, the brush shall be dried for 2 hours and examined for deterioration of the handle base material.

B. Filler strips - Filler strips shall be examined for softening, swelling, or other deleterious change.

C. Setting compound - The setting compound shall be examined for softening, blistering, cracking, or crazing. After drying the brush, the brush shall be suspended by the handle in a holding device, and a group of bristles, 6 mm. (1/4 in.) in diameter, shall be given a tensile pull of 4.5 kg. (10 lbs) away from the ferrule. Repeat this tensile pull test at random places one time for each inch of brush width.

4.4.3 Drop test. Remove bristle (or synthetic filament) and ferrule. Synthetic handles shall be dropped twice from a height of three meters (10 feet) onto a concrete surface and examined for compliance with 3.1.4.

4.4.4 Straightness. The assembled brush head shall be placed so that the bristle is immersed in 60 degree C (140 degree F) water for five (5) minutes. Excess water shall be shaken out. The brush shall be hanged with bristle down to dry completely. The flare shall be measured by aligning the ferrule portion of the brush as shown in figure 1. Flare measurement is limited to the outline of the main body of the brush. Stray bristles shall be disregarded.

4.4.5 Tipping and flagging of filament. The depth of tipping and flagging shall be measured with a rule and a 7-power magnifying glass.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The preservation, packaging, packing, and marking shall be in accordance with PPP-B-1880. The level of packaging and packing shall be A, B, C, or otherwise specified in the contract or order (see 6.2). The marking shall be civil, military, or as specified (see 6.2).

6. Notes

6.1 Intended use. Brushes made with natural hog bristle (Type I) are recommended for application of petroleum based paints and coatings. Brushes made with synthetic filaments (Type II) are recommended for application of water based latex paints and coatings.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement of documents:

(a) Title, number, and date of this specification.
(b) Type, grade, and size required (see 1).
(c) Levels of packaging and packing required (see 5.1).

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(d) Marking required (see 5.1).

6.3 Definitions

6.3.1 Flagging. Flagging is a mechanical process to split filament ends to simulate the natural flag of hog bristle.

6.3.2 Tipping. Tipping is a process of grinding or sanding filament tips to produce a fine, soft, even tip.

PREPARING ACTIVITY:

GSA - FSS

CIVIL AGENCIES COORDINATING:

GSA - FSS

JUS - FPI

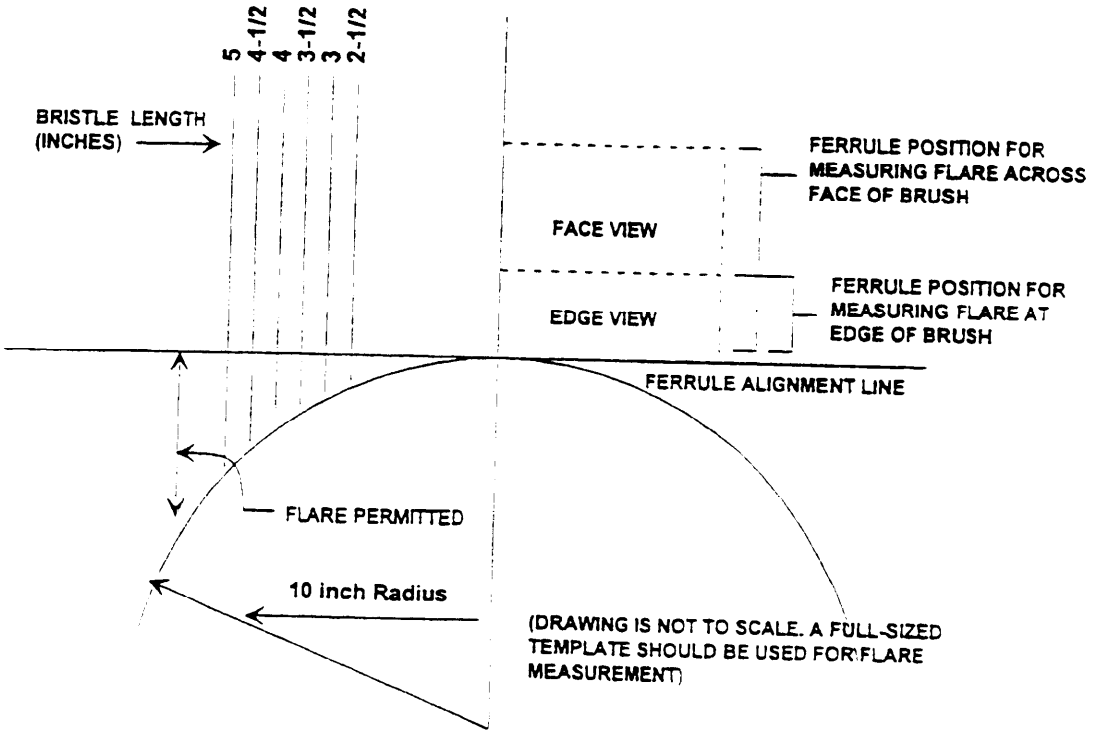


Figure 1. Flare measurement

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